PS-8 run report: meet-in-the-middle pair table against the pairs-and-solve kernels

PDP oracle speedup program, `research/pdp_oracle_speedup_20261011/ps8_mitm/raw/` — 2026-10-18, 18:04–18:32 UTC

1. What ran

`ps8_mitm/run_ps8.sh` (correctness, then the interleaved timing stage), `examples/pdp_speedup` arm `mitm` (`mitm.rs`), one release binary for every arm (SHA-256 in `raw/binary_hashes.txt`), protocol `protocols/PS-8.md` frozen before the first measurement. Host: Intel Xeon 2.10 GHz KVM guest, 4 vCPU, kernel `6.18.44-fc-v139` (the second host class of PS-3); timed runs on CPU 3 through `tools/isolated_bench.py run --cpus 3` (a shared container, not the repository's receipt service). Arms: `baseline`, `baseline2` (A/A), `group-scalar` (PS-2), `bs512` (PS-3), `mitm`. Five rounds, every round runs every arm once.

Runs requested and completed. 150 timed rows requested, 150 completed, 0 marked contended, 0 refused starts. Correctness: 13 `verify-arm` records (`mitm` and `group-scalar` on $l = 8..12$, 80 targets each; `baseline`, `group-scalar`, `mitm` on the 3-target $l = 13$ sample), 0 mismatches.

2. Observations

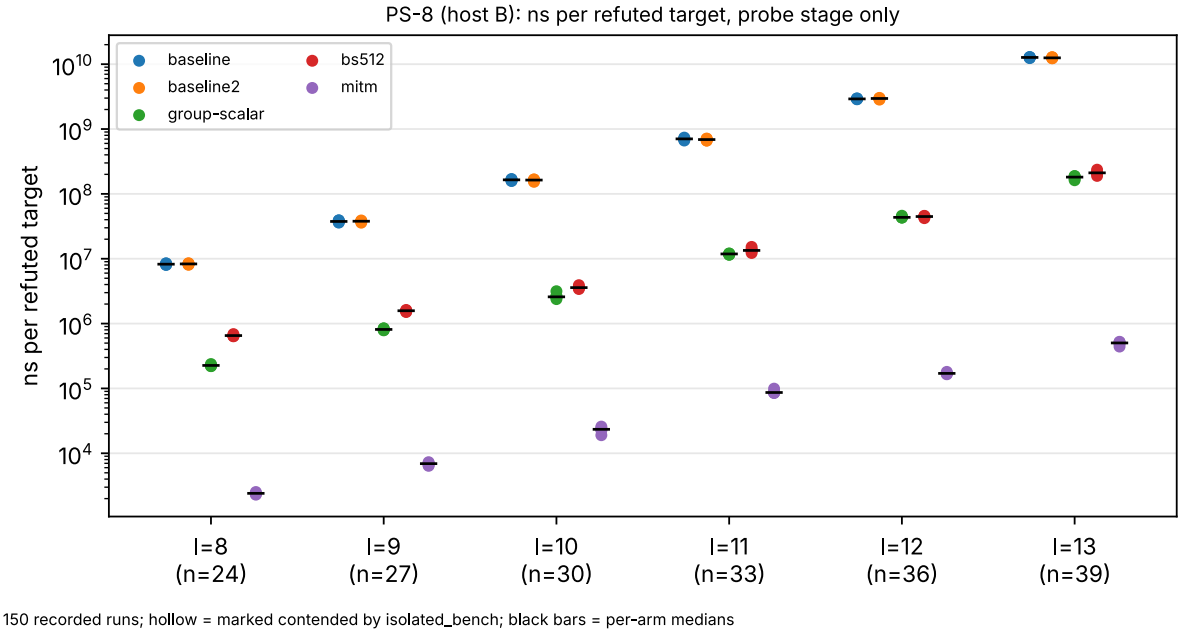


Figure 1: Wall nanoseconds per refuted target by arm and rung, probe stage only (source `raw/times.jsonl`; black bars are per-arm medians; log scale). The table's build is not in these rows.

l	group-scalar ms	bs512 ms	mitm probe µs	mitm build s	keys	MiB	N* vs group-scalar
8	0.226	0.653	2.4	0.0007	16,383	0.4	3.0
9	0.812	1.577	6.9	0.0026	64,008	1.5	3.3
10	2.584	3.587	23.4	0.0126	215,295	6.0	4.9
11	11.854	13.430	86.5	0.0634	1,014,048	24.0	5.4
12	43.455	44.739	170.6	0.2388	3,968,063	96.0	5.5
13	181.363	211.016	502.8	1.4068	16,418,703	384.0	7.8

Medians over five rounds (build: median of the five runs' setup). N^* is derived: build seconds divided by the per-target saving. A/A (baseline against baseline) run-to-run spread: 5.38%, 8.01%, 6.74%, 10.22%, 3.14%, 3.53% at $l = 8..13$.

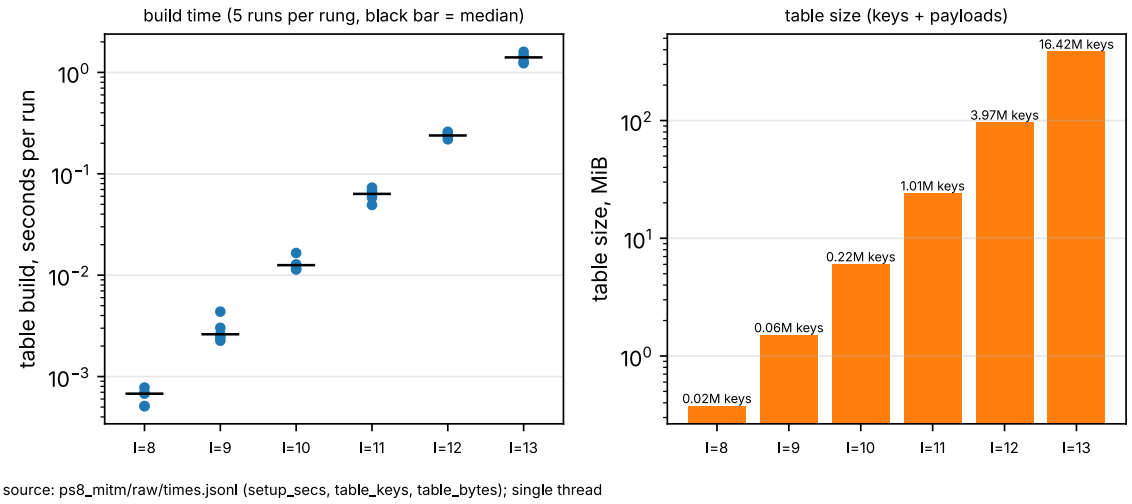


Figure 2: Table build seconds per run (left; five runs per rung, bars are medians) and table size in MiB (right), source `raw/times.jsonl`.

Probe against the scalar group kernel per target (B/A medians): 0.0106, 0.0087, 0.0094, 0.0074, 0.0039, 0.0028 at $l = 8..13$. The hypotheses of the protocol (identical outputs; probe at most 1/10 of the kernel's at $l \geq 10$; break-even at most 100 targets; at most 40 bytes per key) all held on this host.

3. Missing or not done

One host class; the repository's own `ic_boundary::PairTable` was not run on this corpus; no multi-threaded build or probe; nothing about relation yield, linear algebra or the counted unit S (PS-6); nothing for ECC2K-130 (memory grows as $|B|^2$). Wall figures are exploratory in the repository's isolation vocabulary.